



AI Workforce & Automation Analysis

Data Visualization Project

AI Workforce & Automation Dataset Kaggle Dataset Analysis

KAGGLE DATASET

PYTHON · MATPLOTLIB · PANDAS · SEABON

Dataset Overview

Key Variables

-  **salary_usd** — Annual compensation in USD
-  **years_experience** — Professional experience in years
-  **education_required** — Required education level
-  **industry** — Sector of employment
-  **remote_ratio** — Percentage of work performed remotely

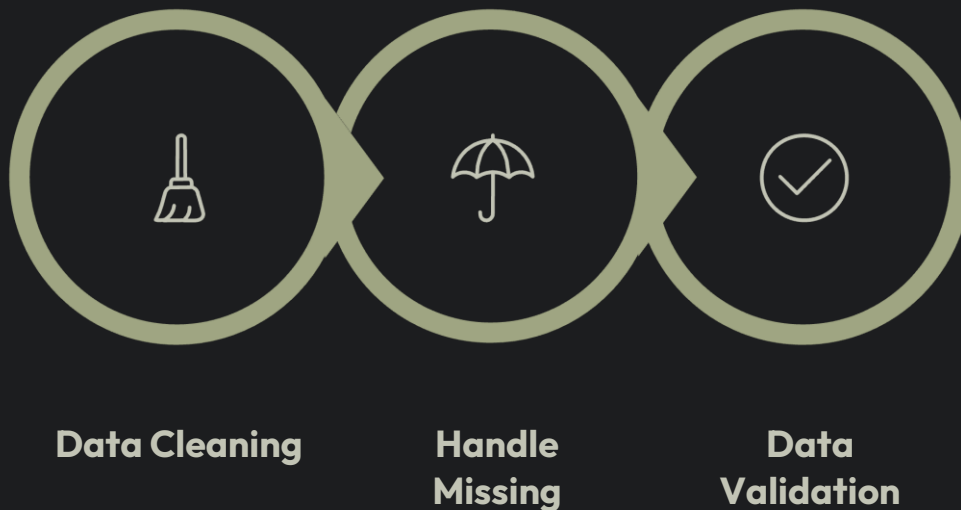
Source & Description

The dataset was obtained from Kaggle and provides insights into the AI job market. It contains information on salaries, experience, education, industries, and workplace characteristics, supporting the analysis of workforce trends and career opportunities in artificial intelligence..



Data Preparation

Before analysis, the dataset underwent a structured preprocessing pipeline to ensure data quality and analytical reliability. This process addressed common data issues such as missing values, inconsistent formatting, and outliers that could skew visualizations and statistical findings.



This three-stage pipeline ensured that all subsequent visualizations and statistical analyses were based on clean, validated data, reducing the risk of misleading patterns or erroneous conclusions.

Dataset Summary: 15,000 records × 20 features, with all variables validated and no missing values detected.

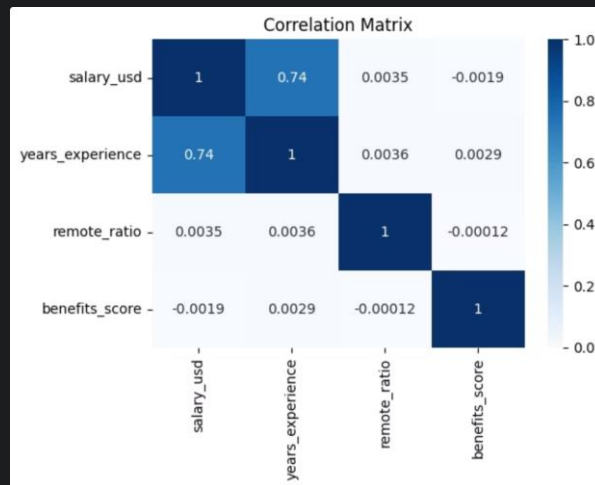
Correlation Analysis

Interpreting the Heatmap

Strongest Relationship: Years of experience and salary show a strong positive correlation of 0.74, confirming experience is the primary driver of higher compensation.

Weak Correlations: Other factors like remote work ratio (remote_ratio) and benefits (benefits_score) show almost zero linear correlation with salary.

Next Steps: These insights guide our next visualizations to focus deeper on analyzing how salary changes across different experience levels

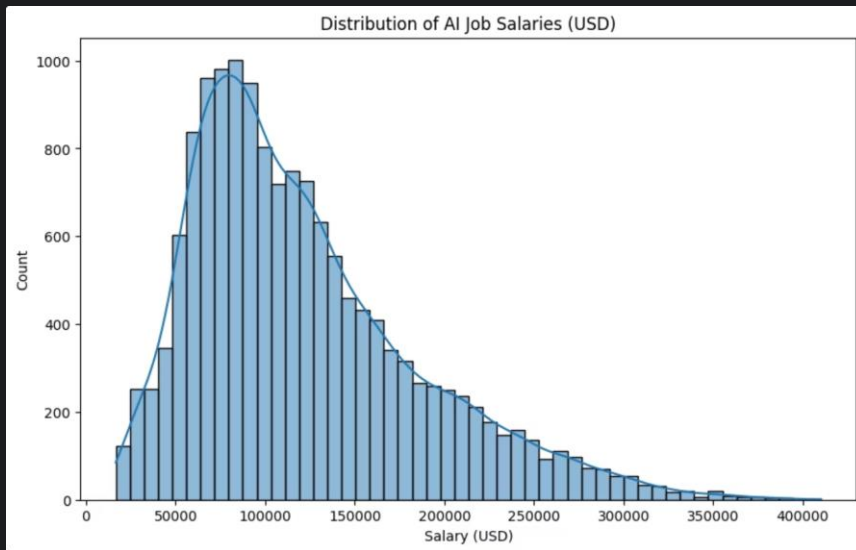


- ③ Higher correlation values indicate stronger linear relationships between variables.

Salary Distribution

Key Observations

The salary distribution is right-skewed, with the majority of AI professionals earning between \$70k and \$110k annually. The peak concentration in the \$70k–\$90k band suggests a strong mid-level workforce, while the sharp drop beyond \$130k indicates that senior and specialized roles are considerably fewer. This distribution pattern is typical of technology labor markets, where a large base of mid-career professionals is supported by a smaller cohort of high-earning experts and leadership roles.



VISUALIZATION 2

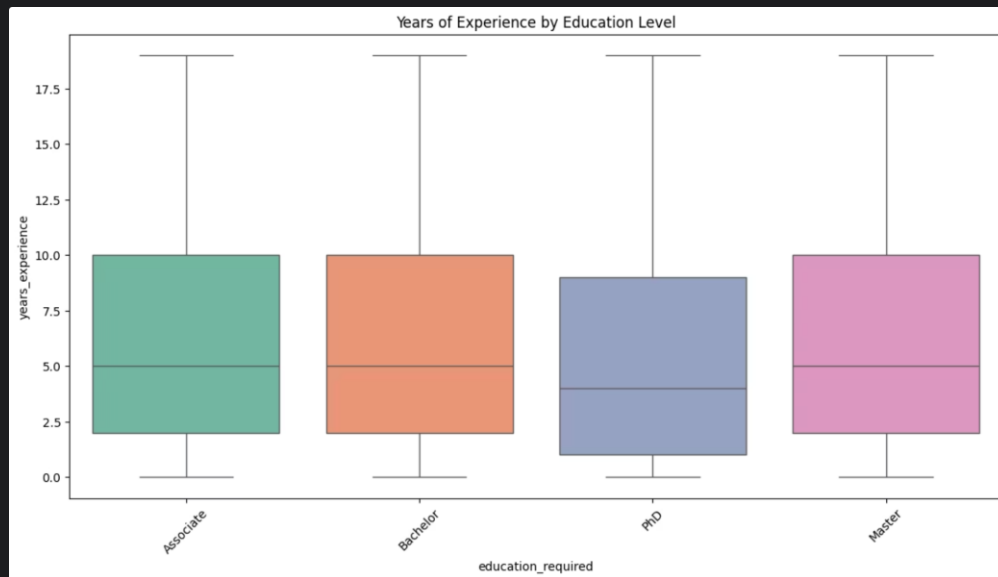
Experience by Education Level

What the Data Shows

Balanced Distribution: The median years of experience remain highly consistent (around 6.4 years) across all education categories (Associate, Bachelor, Master, and PhD).

Uniform Range: For all degree types, the overall experience spans from 0 to a maximum of 19 years.

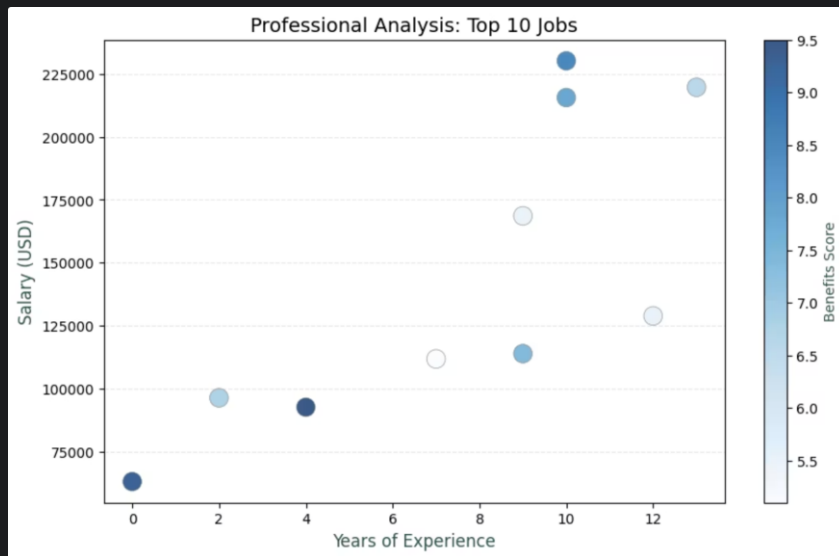
Key Takeaway: Higher academic degrees (like a PhD) do not automatically imply a higher average of years of experience within this dataset.



Salary vs Years of Experience

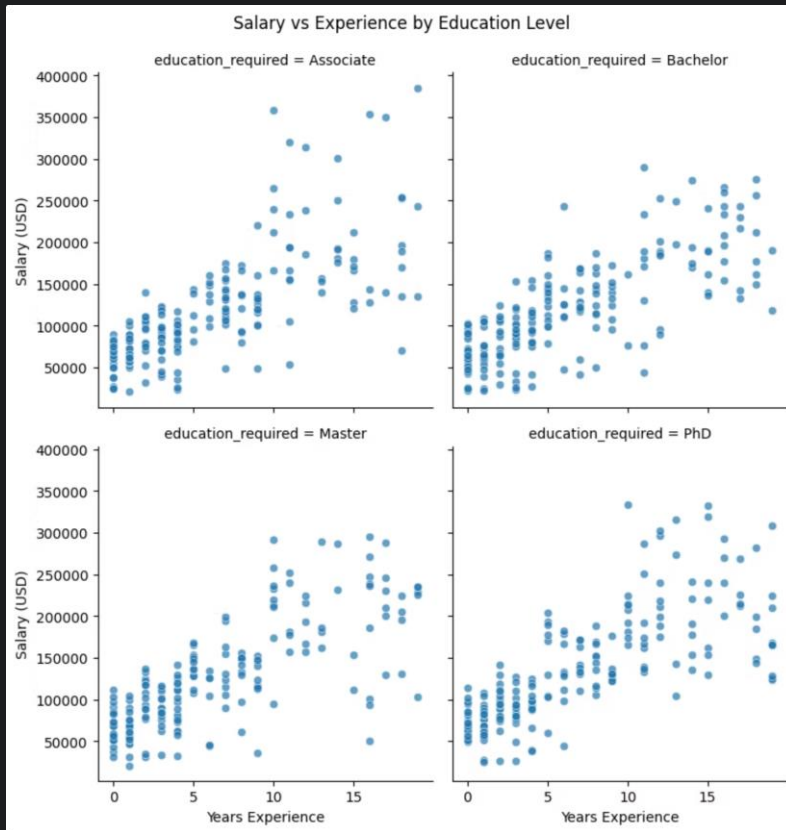
Relationship Analysis

The scatter plot reveals a clear positive relationship between years of experience and salary in the AI workforce, while factoring in overall employment perks. Compensation grows steadily with experience; early-career professionals (0–4 years) cluster between \$75k and \$100k. Mid-career and senior professionals (9+ years) see exponential salary growth, with top positions peaking between \$200k and \$225k. Additionally, the data points are color-coded by Benefits Score (ranging from 5.5 to 9.5). The visualization demonstrates that higher-salaried positions for experienced professionals not only offer top-tier compensation but also tend to align with maximum benefits scores (represented by the darker blue shades), indicating that total compensation packages become significantly more lucrative over time.



Salary Growth with Experience by Education Level

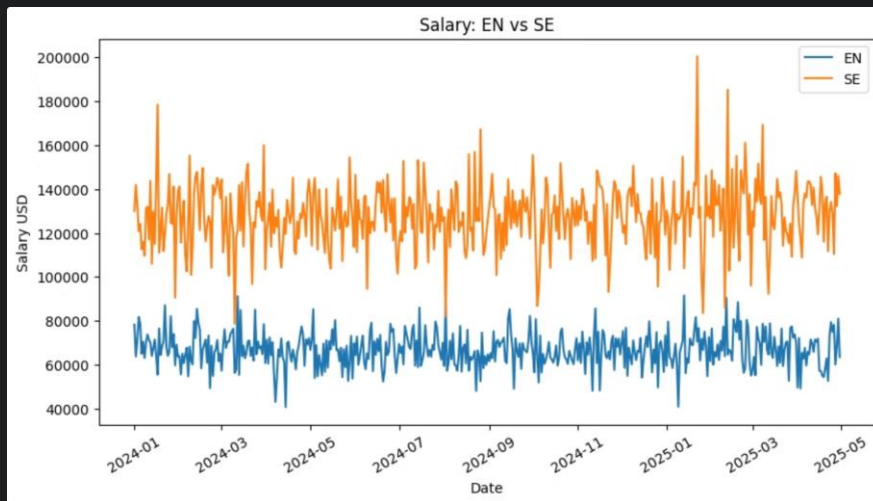
The faceted scatter plots illustrate how salary grows with years of experience across four distinct education levels required for AI roles. Interestingly, the data shows substantial salary variance regardless of the degree held. Across all categories—Associate, Bachelor, Master, and PhD—early career salaries (0-5 years) consistently cluster under \$150k. As experience progresses beyond 10 years, significant upward mobility is visible in all groups. Notably, the Associate and Bachelor tracks display several high-earning outliers reaching between \$300k and \$400k. While postgraduate degrees (Master's and PhDs) guarantee strong baseline compensation steady above \$150k in mid-to-late career stages, they show a wider spread of data points, indicating that experience and specialized skills in the AI labor market can drive executive-level salaries even without a doctorate.



VISUALIZATION 5

Advanced Visualization: Salary Trends Over Time (EN vs SE)

The time-series plot shows a clear and continuous gap between entry-level (EN) and senior-level (SE) AI salaries from 2024 to mid-2025: Entry-Level (EN): Salaries (blue line) are highly stable, tightly clustering between \$60k and \$80k. Senior-Level (SE): Salaries (orange line) maintain a much higher baseline around \$120k but exhibit high volatility with sharp peaks reaching up to \$200k.



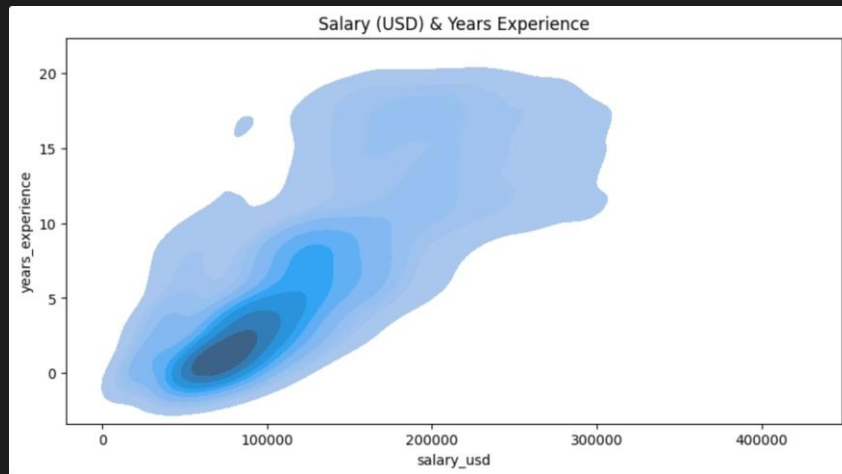
VISUALIZATION 6

Bivariate Density: Experience vs Compensation

Experience vs Compensation Early Career (0-5 Years):
Heavy cluster under \$100k, representing the entry baseline.

Mid to Late Career: Distribution broadens significantly, opening access to diverse salary brackets.

The Insight: More experience directly drives salary progression and unlocks executive-level tiers.



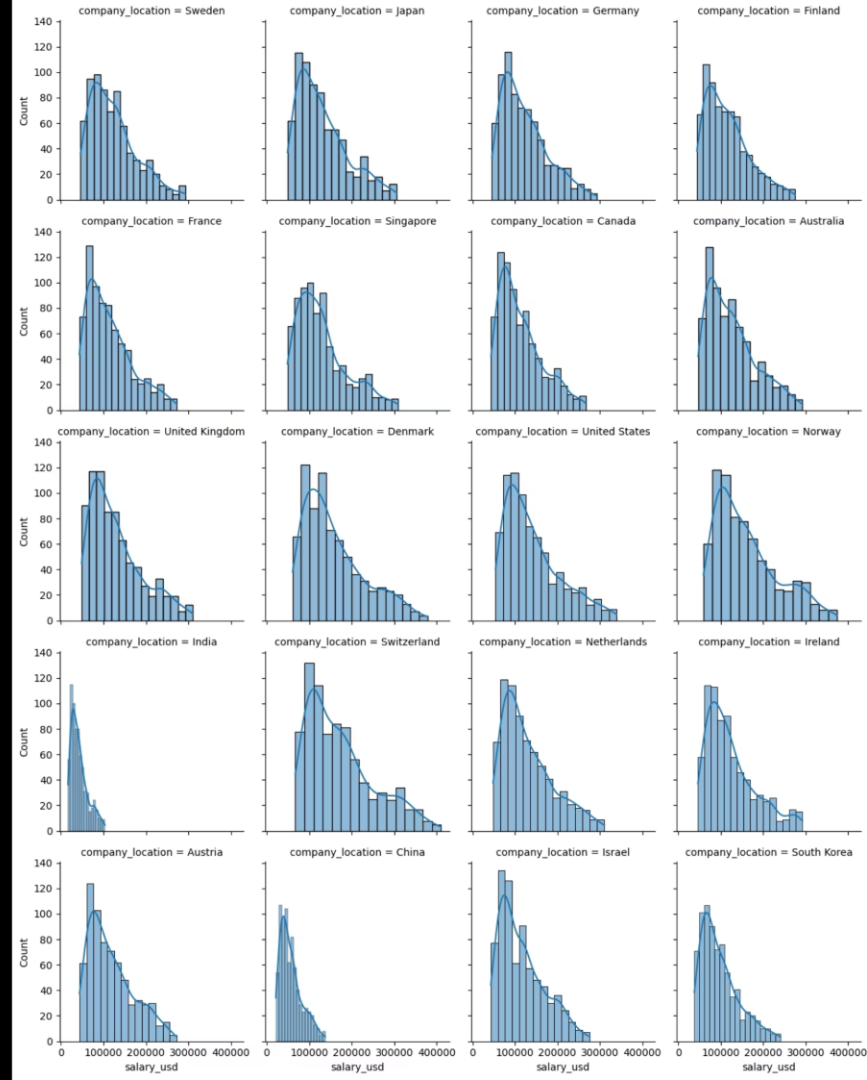
VISUALIZATION 7

Global Salary Distribution by Company Location

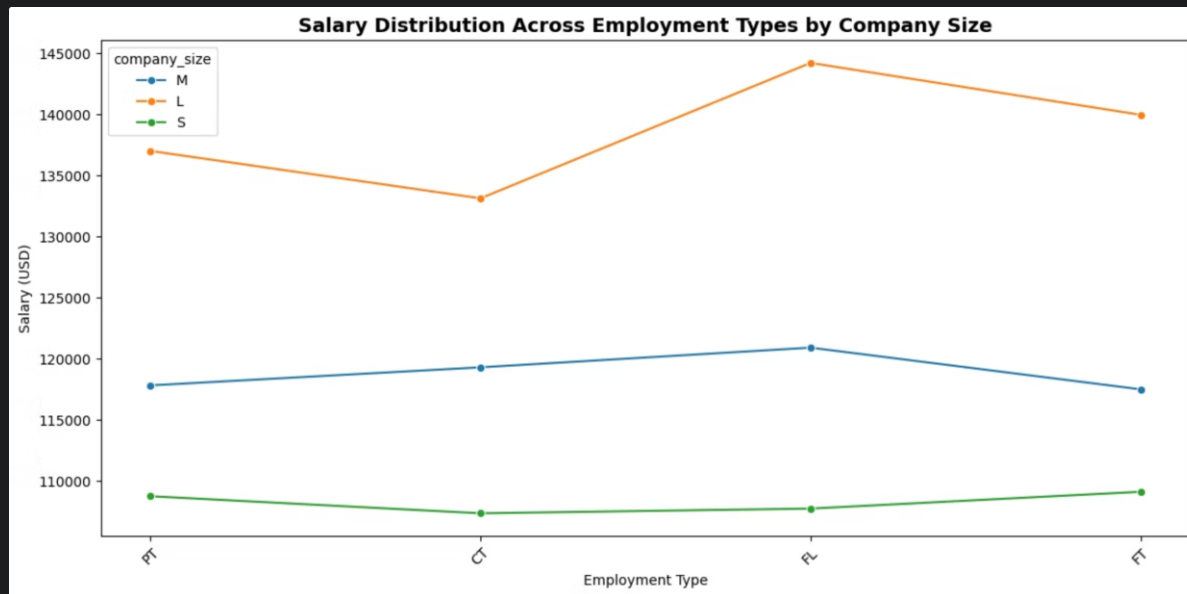
Core Focus: Shows the distribution of AI salaries (salary_usd) across different company locations.

Distribution Shape: The data is right-skewed across almost all countries (e.g., Germany, Japan, Sweden, Finland).

Key Takeaway: This skewness indicates that the majority of AI professionals earn within the low-to-mid range, while only a small percentage reach the highest salary brackets globally.



Salary Distribution Across Employment Types by Company Size



The Code

Function: Plots Salary against Employment Type, colored by Company Size.

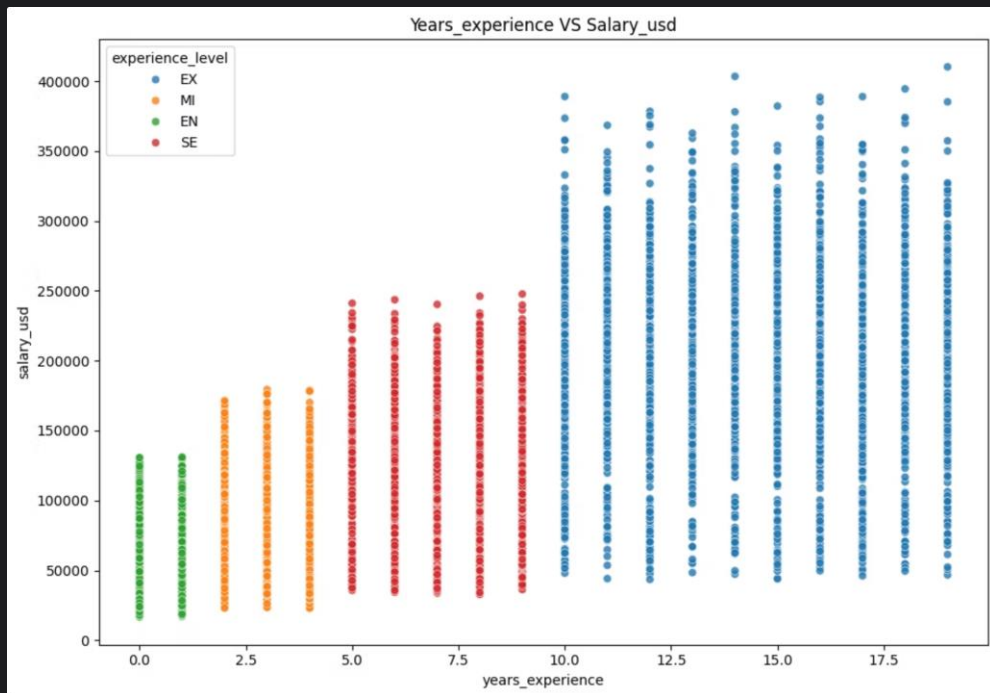
Note: The warning asks you to replace `ci=None` with `errorbar=None` because it is deprecated. The Plot Large Companies (L): Pay the highest salaries by far, varying based on employment type (peaking at Freelance FL).

Medium (M) & Small (S) Companies: Pay lower salaries, and the employment type barely changes the amount (lines are flat).

Salary Trend Analysis in the AI Workforce

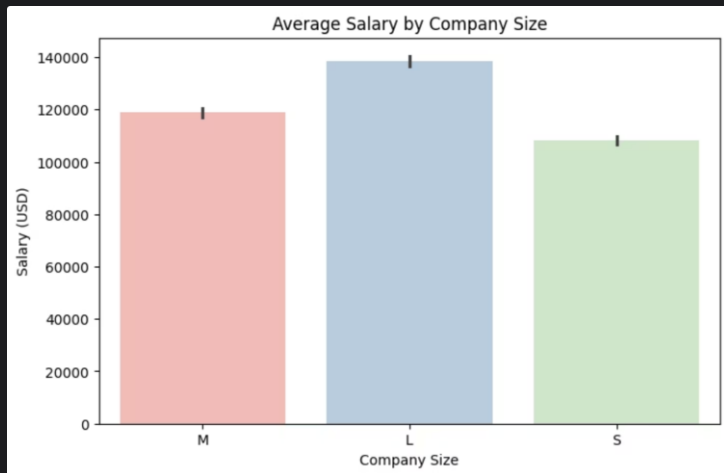
This analysis aims to uncover the dynamics of compensation within the burgeoning AI industry, specifically focusing on how salary structures evolve with professional experience.

- To comprehensively examine salary progression across different experience levels.
 - To identify key determinants influencing remuneration in various AI roles.
- To furnish data-driven insights pertinent to career development and talent acquisition strategies.



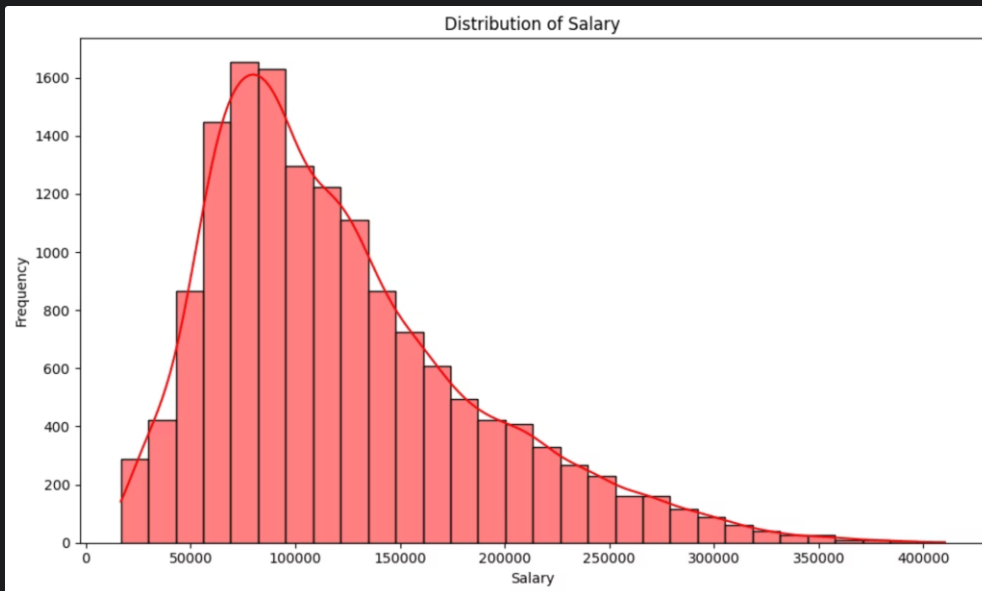
Introduction: Organisational Scale & Compensation

- Main Objective: The slide shows how company size impacts AI workforce salaries.
- Large Companies (L): Offer the highest average salary, reaching close to \$140,000.
- Medium Companies (M): Come in second place, with an average salary of around \$120,000. Small Companies (S): Offer the lowest average salary, hovering around \$110,000.



Introduction to Salary Distribution

- The Code: Uses a Histogram with a KDE curve (`sns.histplot`) to show how often different salary amounts appear in the dataset.
- The Chart: Shows that the majority of salaries are concentrated/clustered in the higher range (around \$130,000 to \$160,000), with fewer people earning very low or very high amounts.

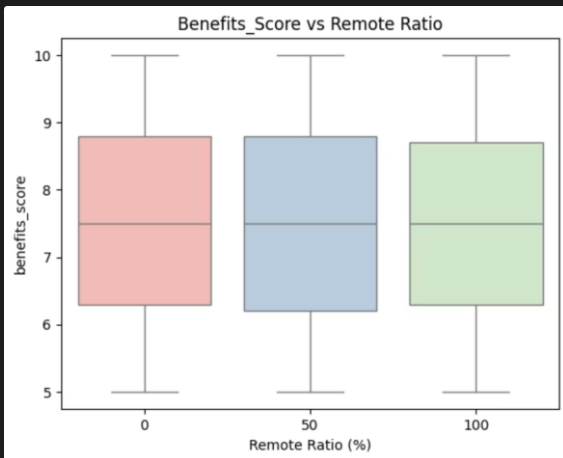


OUTLINE

Introduction to Benefits & Remote Work

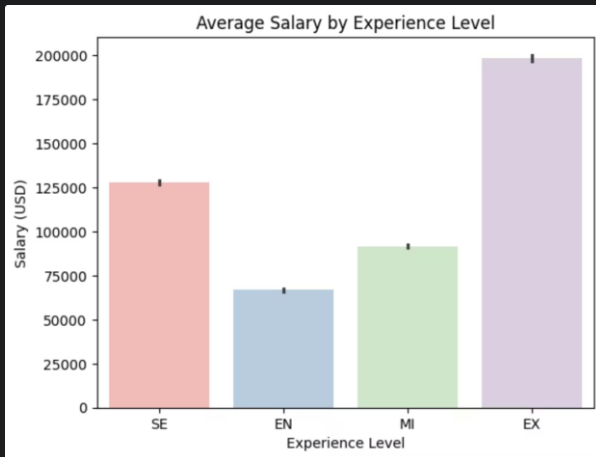
The Code: Uses a Boxplot (`sns.boxplot`) to compare how the benefits score changes depending on the percentage of remote work (0%, 50%, or 100%).

The Chart: Shows that the median and overall distribution of the benefits score are very similar across all remote work ratios, meaning working remotely doesn't significantly change the employee benefits score.



Introduction to Career Progression & Compensation

- The Code: Uses a Bar Plot (`sns.barplot`) to display and compare the average salary (in USD) across four defined experience levels (`levels = ['SE', 'EN', 'MI', 'EX']`).
- The Chart: Visualizes the clear upward trend in compensation, demonstrating how average salary directly scales up as an employee moves from entry-level positions toward executive roles.



Key Findings & Conclusion

Experience Drives Salary

Years of experience is the strongest predictor of compensation, with a clear positive correlation (~ 0.72) confirmed across all education levels.

Education Accelerates Growth

Advanced degrees consistently command higher salaries, with PhD holders earning up to \$70k more than Bachelor's holders at senior levels.

Mid-Range Concentration

Most AI professionals cluster in the \$70k–\$110k salary band, reflecting a strong mid-level workforce with fewer senior roles.

Implications

Organizations should invest in experience-based progression pathways and support advanced education to attract and retain top AI talent.